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Curriculum Vitae

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Research Focus

Multimodal machine learning for multi-scale biomedical data fusion

A. Personal Statement

I am an integrated M.S./Ph.D. student in the Department of Medical Science at Yonsei University College of Medicine in South Korea, pursuing the Genomics and Data Sciences track. My research focuses on multimodal machine learning for multi-scale biomedical data integration. I am particularly interested in developing computational frameworks that link molecular profiles with imaging and tissue-level phenotypes to improve disease stratification, prediction, and biological understanding.

More broadly, my goal is to build translational AI approaches that integrate high-dimensional molecular and clinical data to uncover biologically meaningful mechanisms and support precision medicine. Grounded in my clinical training in nursing, I am particularly motivated to pursue research that bridges computational modeling with patient-relevant biomedical applications.

B. Education

Mar 2023 – Present: Yonsei University College of Medicine, Seoul, Korea
Integrated M.S./Ph.D. student, Department of Medical Science
(Supervisor: Prof. Sungsoon Fang)

Sep 2021 – Feb 2023: Yonsei University College of Life Science and Biotechnology, Seoul, Korea
B.E., Department of Biotechnology (Double major)

Mar 2017 – Feb 2023: Yonsei University College of Nursing, Seoul, Korea
B.S.N., Department of Nursing

C. Peer-reviewed Publications

1) **Lee CM**[†], Kim J[†], Song J, Kim AS, Jo S, Hwang N, Jeong JW, Kim M, Shin SJ, Fang S*, Yoon BK* (2026) Endothelial cell senescence shapes T cell activity in late-stage of Chronic Obstructive Pulmonary Disease. ***Cell Death Discovery*** **(IF 10.4)** (link)
2) **Lee CM**[†], Kim AS[†], Kim M, Jeong JW, Jo S, Hwang N, Fang S* (2025) Dynamics of T cell-mediated immune signaling network during pathogenesis of chronic obstructive pulmonary disease. ***Yonsei Medical Journal*** 66(6):354-365 **(IF 2.8)** (link)
3) **Lee CM**, Hwang Y, Kim M, Park YC, Kim H, Fang S* (2024) PHGDH: A novel therapeutic target in cancer. ***Experimental & Molecular Medicine*** 56(7):1513-1522 **(IF 12.9)** (link)
4) **Lee CM**[†], Hwang Y[†], Jeong JW, Kim M, Lee J, Bae SJ, Ahn SG*, Fang S* (2024) BRCA1 mutation promotes sprouting angiogenesis in inflammatory cancer-associated fibroblasts of Triple-negative breast cancer. ***Cell Death Discovery*** 10:5 ***(CDD press: Monthly Reader's Choice)*** **(IF 7.0)** (link)
5) **Lee CM**, Fang S* (2023) Fat Biology in TNBC: Immune Regulation, Fibrosis, and Senescence. ***Journal of Obesity & Metabolic Syndrome*** 32(4):312-321 **(IF 7.9)** (link)
6) Kim K, Fang S, Hong SH, Cho H, Leblanc M, Jacinto S, Schnable B, Atkins A, Yu RT, Liddle C, Truitt ML, Fan W, **Lee CM**, Downes M*, Evans RM* (2026) SMRT regulation of nuclear receptors orchestrates bile acid homeostasis. ***Experimental & Molecular Medicine*** **(IF 17.5)** (link)

7) Jo S, Ok T, **Lee CM**, Jeong JW, Lee YJ, Shin SJ, Fang S, Lee K*, Yoon BK* (2025) Spatial immune landscape of large artery atherosclerotic and cardioembolic thrombus retrieved from mechanical thrombectomy. ***The FASEB Journal*** 39(23): e71283 **(IF 4.2)** (link)
8) Hwang Y†, **Lee CM**†, Sung M, Park Y-C, Yoon BK*, Fang S*. (2026) One-carbon metabolism-high fibroblast predicts poor prognosis in colorectal cancer. ***npj precision oncology*** **(IF 9.9)** *(In revision)*
9) **Lee CM**, Kim KY, Fang S*, Kim B-H*. SNP-derived pathway burden identifies functional connectivity biomarkers of Alzheimer's disease progression. *(submitted)*

D. Techniques

1) **Computational Omics and Spatial Biology**
   - Bulk RNA-seq, scRNA-seq, snATAC-seq
   - Spatial transcriptomics: CosMx, GeoMx, Visium, Xenium
2) **Multimodal Modeling and Imaging-Linked Prediction**
   - Multimodal self-supervised learning
   - Neuroimaging-genetics integration
   - Predictive modeling linking molecular, imaging, and phenotype-level data
   - UK Biobank large-scale biomarker modeling
3) **Experimental Methods**
   - Cell culture, transfection, gene silencing (siRNA)
   - Western blotting and protein expression assays
4) **Animal Models**
   - Mouse disease models
   - AAV administration
   - Intratracheal instillation
   - Knockout mouse line maintenance

E. Research Interests

- Multimodal AI for multi-scale biomedical data fusion
- Multi-omics integration and spatial biology
- Interpretable machine learning for biomedical decision support
- Precision medicine through molecular and imaging data integration

F. Manuscripts in preparation

1) **Lee CM**, Fang S*, Kim B-H*. Foundation model adaptation for EEG-based brain phenotyping. *(In preparation)*
2) Hwang Y†, **Lee CM**†, Yoon BK, Fang S*, Lee YH*. Endothelial cell-mediated interaction patterns of syndecan family members differ between MVI-negative and MVI-positive hepatocellular carcinoma. *(In preparation)*
3) **Lee CM**†, Yoon BK*, Fang S*. Multi-omics characterization of the liver-ileum-eWAT axis reveals inter-organ signaling following RYGB surgery in a rat model. *(In preparation)*
4) Sung M†, **Lee CM**†, Fang S*. One-carbon metabolism remodeling across the MASLD-MASH continuum reveals candidate therapeutic targets. *(In preparation)*
5) **Lee CM**, Fang S*, Jung M*, Jhee JH*. Stage-specific remodeling of the periglomerular immune microenvironment in kidney injury revealed by CosMx spatial transcriptomics. *(In preparation)*
6) **Lee CM**†, Lee YJ†, Song J, Sung M, Chung HW, Fang S*, Yoon BK*. Sphingosine-1-phosphate signaling contributes to fibroblast-associated remodeling in idiopathic pulmonary fibrosis. *(In preparation)*

7) **Lee CM**[†], Lee YJ[†], Song J, Fang S*, Yoon BK*. Fibroblast-specific pathway remodeling linked to glycoprotein acetyls identifies candidate therapeutic targets in idiopathic pulmonary fibrosis. *(In preparation)*

G. Research Experiences

1) **Graduate Researcher**

Mar 2023 – Present: Yonsei University College of Medicine, Seoul, Korea
Laboratory of Dynamics in Cellular Energetics (Supervisor: Prof. Sungsoon Fang)

- Conduct multi-omics and spatial transcriptomics analyses for therapeutic target discovery in metabolic, fibrotic, and cancer-related diseases
- Develop computational pipelines for integrating public and in-house multi-omics datasets to characterize disease heterogeneity and identify clinically actionable molecular programs
- Build multimodal predictive models linking molecular profiles with imaging or phenotype-level information for disease stratification

2) **Undergraduate Research Intern**

Jan 2020 – Feb 2021: Department of Biotechnology, Yonsei University, Seoul, Korea
Laboratory of Medicinal Chemistry (Supervisor: Prof. Gyoonhee Han)

- Synthesized PAR2 antagonists and characterized compounds using chromatographic separation and NMR analysis
- Performed molecular docking analysis of FLT3 inhibitors using Discovery Studio for AML drug design

Jan 2020 – Jun 2020: College of Nursing, Yonsei University, Seoul, Korea
Mar 2021 – Jun 2022 Laboratory of Behavioral and Physiological Health (Supervisor: Prof. Hyangkyu Lee)

- Investigated dietary regulation of gene expression using Western blot analysis
- Analyzed exosome miRNA to recognize the cause of delirium via NTA

Jul 2022 – Feb 2023: Yonsei University College of Medicine, Seoul, Korea
Laboratory of Dynamics in Cellular Energetics (Supervisor: Prof. Sungsoon Fang)

H. Teaching & Academic Service

1) **LT (learn-by-teaching) lecturer**

Mar 2025 – Present: Yonsei University College of Medicine, Seoul, Korea
'Joint Seminar' (MES6060.01-00)

2) **Graduate Student Representative, Department of Medical Science**

Mar 2026 – Present: Yonsei University College of Medicine, Seoul, Korea

3) **Ad hoc Peer Reviewer**

2026: *Clinical and Translational Discovery; Clinical and Translational Medicine; International Journal of Chronic Obstructive Pulmonary Disease; Journal of Multidisciplinary Healthcare; Medicinal Research Reviews*

I. Honors, Awards & Selected Presentation

- **2026 KALAS Student Research Award**
  Korean Association for Laboratory Animal Science (KALAS), 2026
- **Outstanding Poster Award, 3rd Place**
  8th KMPC-KSBMB Energy Metabolism Division Camp, 2026
- **Selected Oral Presenter, Young Scientists Session**
  8th KMPC-KSBMB Energy Metabolism Division Camp, 2026

J. Patent & License

1) **Korea patent publication**

: Hospital Fall Prevention Wheelchair (application number: 10-2023-0017198)

2) **License**
: Registered Nurse (Korean Ministry of Health and Welfare)
: NCLEX-RN – In progress

K. References

1) **Prof. Sungsoon Fang, Ph.D.** *(Ph.D. advisor)*
Dept. of Biomedical Science, Yonsei University College of Medicine, Seoul, Korea
sfang@yuhs.ac, https://sites.google.com/view/fang-lab/home

2) **Prof. Bo Kyung Yoon, MD, Ph.D.** *(co-advisor)*
Dept. of Microbiology and Immunology, Yonsei University College of Medicine, Seoul, Korea
yoonbbk89@yuhs.ac, https://sites.google.com/view/immunometabolism-lab

3) **Prof. Byung-Hoon Kim, MD, Ph.D.** *(co-advisor)*
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